Project Assignment 0

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CSE5825 - BML

Description of Data

Reddit is a popular, anonymous online forum self-described as "the most real place on the internet," where users can create posts, write comments, and vote on submissions using upvotes and downvotes. The content on Reddit ranges from news articles to niche interests to anonymous venting posts, and is organized into smaller subforums called subreddits. Each subreddit is a microcosm with its own rules, moderation, and community culture.

The dataset for this project is the open-index/arctic dataset on Huggingface, consisting of all posts and comments on Reddit since 2005 from every publicly available subreddit. The dataset, which is 1.3 TB, has 15.7 billion total entries, with 12.9 billion comments and 2.8 billion posts. Posts and comments are stored as separate datasets and split into Parquet shards.

Data fields for both posts and comments include the Reddit-generated ID, the author, subreddit to which it was posted, content body (and title for posts), timestamp of creation, and score (number of upvotes – can be positive or negative).

Comments (`data/comments/YYYY/MM/NNN.parquet`)

Column	Type	Description
`id`	VARCHAR	Reddit's base-36 comment ID
`author`	VARCHAR	Username. `[deleted]` if account was removed
`subreddit`	VARCHAR	Subreddit name (no `r/` prefix)
`body`	VARCHAR	Comment text in Markdown
`score`	BIGINT	Net upvotes at time of archival
`created_utc`	BIGINT	Unix timestamp
`created_at`	TIMESTAMP	Derived from `created_utc`
`body_length`	BIGINT	Character count of `body`
`link_id`	VARCHAR	Parent submission ID (`t3_...` format)
`parent_id`	VARCHAR	Parent comment or submission ID
`distinguished`	VARCHAR	`moderator`, `admin`, or null
`author_flair_text`	VARCHAR	Author's flair in this subreddit

Submissions (`data/submissions/YYYY/MM/NNN.parquet`)

Column	Type	Description
`id`	VARCHAR	Reddit's base-36 submission ID
`author`	VARCHAR	Username of the poster
`subreddit`	VARCHAR	Subreddit name
`title`	VARCHAR	Post title
`selftext`	VARCHAR	Post body for text posts (empty for link posts)
`score`	BIGINT	Net upvotes at time of archival
`created_utc`	BIGINT	Unix timestamp
`created_at`	TIMESTAMP	Derived from `created_utc`
`title_length`	BIGINT	Character count of `title`
`num_comments`	BIGINT	Comment count on this post
`url`	VARCHAR	External URL for link posts, permalink for text posts
`over_18`	BOOLEAN	NSFW flag
`link_flair_text`	VARCHAR	Post flair text
`author_flair_text`	VARCHAR	Author's flair

Hypotheses

The Reddit Explore Page provides a comprehensive list of the types of subreddits on the site, which can further be generalized into the following categories:

News and Politics	Discussion and Support
Technology and Science	Media and Entertainment
Business and Finance	Lifestyle, Hobbies and Living
Education and Career	

Using these categories, the following three hypotheses may be supported by the data:

1. Identifying Manufactured Popularity/Astroturfing
   Do posts in political, news, technology, and finance subreddits tend to have more abnormal engagement than posts in more casual subreddits such as hobby and general discussion subreddits?

2. Changes in Discourse on AI Over Time
   Do technology subreddits display an increase in negative or skeptical language towards AI over time?

3. Detecting AI-Generated Content Across Subreddits
   Do certain subreddits contain a higher proportion of posts and comments that appear to be written by AI? For example, posts with little sentence variety, repetitive phrasing, and overuse of buzzwords.